

Implementation Report: Physics-Informed Variational Autoencoder (PIVAE) for Airfoil Parameterization

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1 Introduction

This report documents the implementation of a Physics-Informed Variational Autoencoder (PIVAE) designed for the generative design and analysis of airfoil geometries. The architecture adopts the airfoil generator (AG) methodology proposed by *Kang et al. (2025)*, which addresses the limitations of traditional generative models regarding feasibility and intuitiveness. The system utilizes a split-network architecture to decouple thickness and camber distributions, ensuring topological validity (non-intersection) and providing direct control over aerodynamic features.

2 Methodology Overview

The overall approach treats airfoil parameterization as a dimensionality reduction problem where high-dimensional coordinates are compressed into a low-dimensional, physically interpretable latent space.

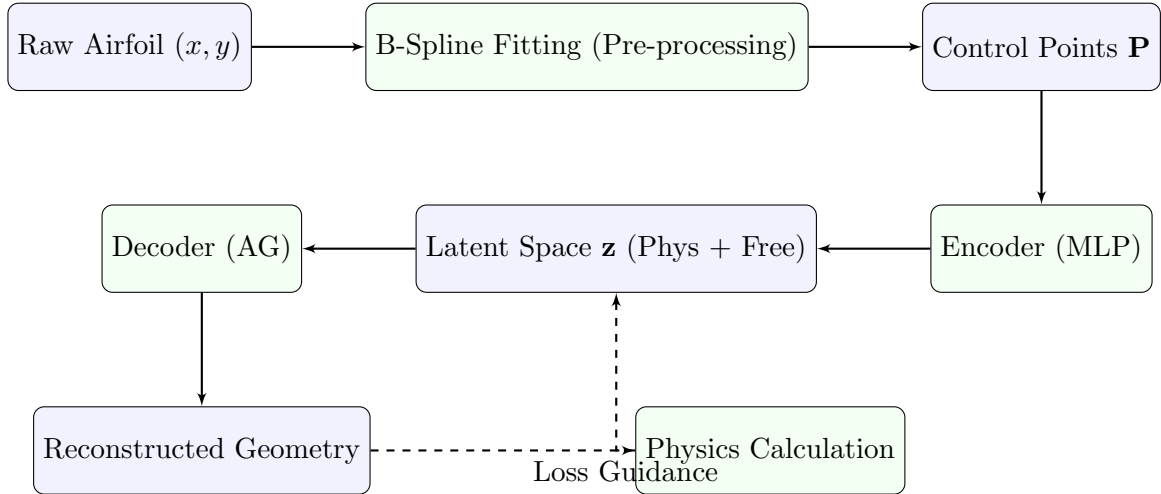


Figure 1: Overview of the PIVAE approach. Raw data is converted to control points using a clustered topology, encoded into a disentangled latent space, and reconstructed via a physics-constrained decoder.

One core innovation of this PIVAE is the dual-alignment strategy that bridges the gap between the Encoder’s latent space and the Decoder’s physical output.

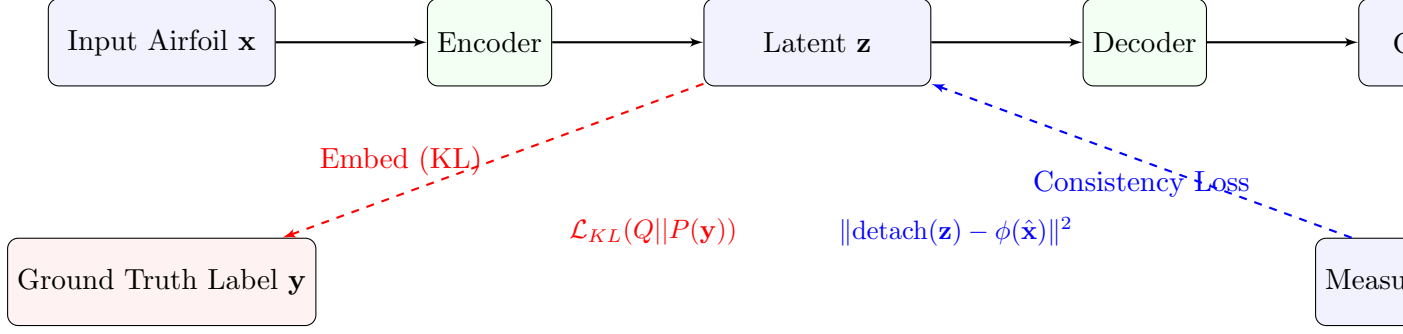


Figure 2: The "Bridge" Strategy. The latent space is anchored from two sides: (1) The Encoder is forced to place latents at the correct physical label coordinates via KL divergence. (2) The Decoder is forced to respect these latents via a Consistency Loss that measures the generated geometry.

3 Geometric Parameterization

3.1 Decomposition and B-Spline Representation

To guarantee geometric feasibility, the airfoil is not represented as a single coordinate loop but is decomposed into two independent curves defined over the normalized chord $x \in [0, 1]$:

1. **Thickness Distribution** ($T(x)$): Symmetric thickness added to the mean line.
2. **Camber Distribution** ($C(x)$): The curvature of the mean line.

The physical upper (y_u) and lower (y_l) surfaces are derived as:

$$y_u(x) = C(x) + \frac{1}{2}T(x), \quad y_l(x) = C(x) - \frac{1}{2}T(x) \quad (1)$$

3.2 Clustered Knot Vector Topology

Standard uniform B-splines often struggle to capture the high curvature gradients at the Leading Edge (LE), leading to "sharp nose" artifacts. To resolve this, we employ a **Clustered Knot Vector** where the knot density follows a power law ($t \sim x^2$):

$$\text{Knots} = \{0, 0, 0, 0, 0.01, 0.04, 0.09, \dots, 1, 1, 1, 1\} \quad (2)$$

This allocates a higher density of control points (P_0, P_1, P_2) to the first 5% of the chord, providing the necessary resolution to model a perfectly round LE radius without increasing the total number of parameters ($N = 16$).

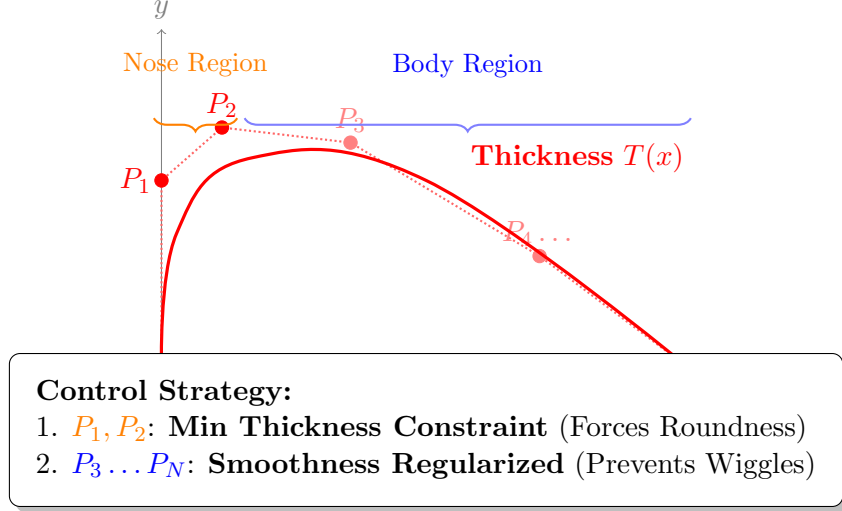


Figure 3: Clustered Control Point Parameterization. Points P_1 and P_2 are packed near the leading edge to control the nose radius. The loss function treats these zones differently to avoid conflicting objectives.

3.3 Preprocessing: Constrained Fitting

We convert raw coordinate data into control points using a constrained least squares algorithm that respects the $P_0 = (0, 0)$ constraint while leveraging the clustered basis matrix $\mathbf{M}_{clust}$:

$$\min_{\mathbf{P}_{free}} \|\mathbf{M}_{clust}\mathbf{P} - \mathbf{Y}_{target}\|^2 + \lambda \|\mathbf{D}^2\mathbf{P}\|^2 \quad (3)$$

4 Decoder Process Scheme

The decoder translates the latent vector $\mathbf{z}$ back into control points. To ensure physical validity (e.g., non-negative thickness and a round nose), we apply region-specific activation functions.

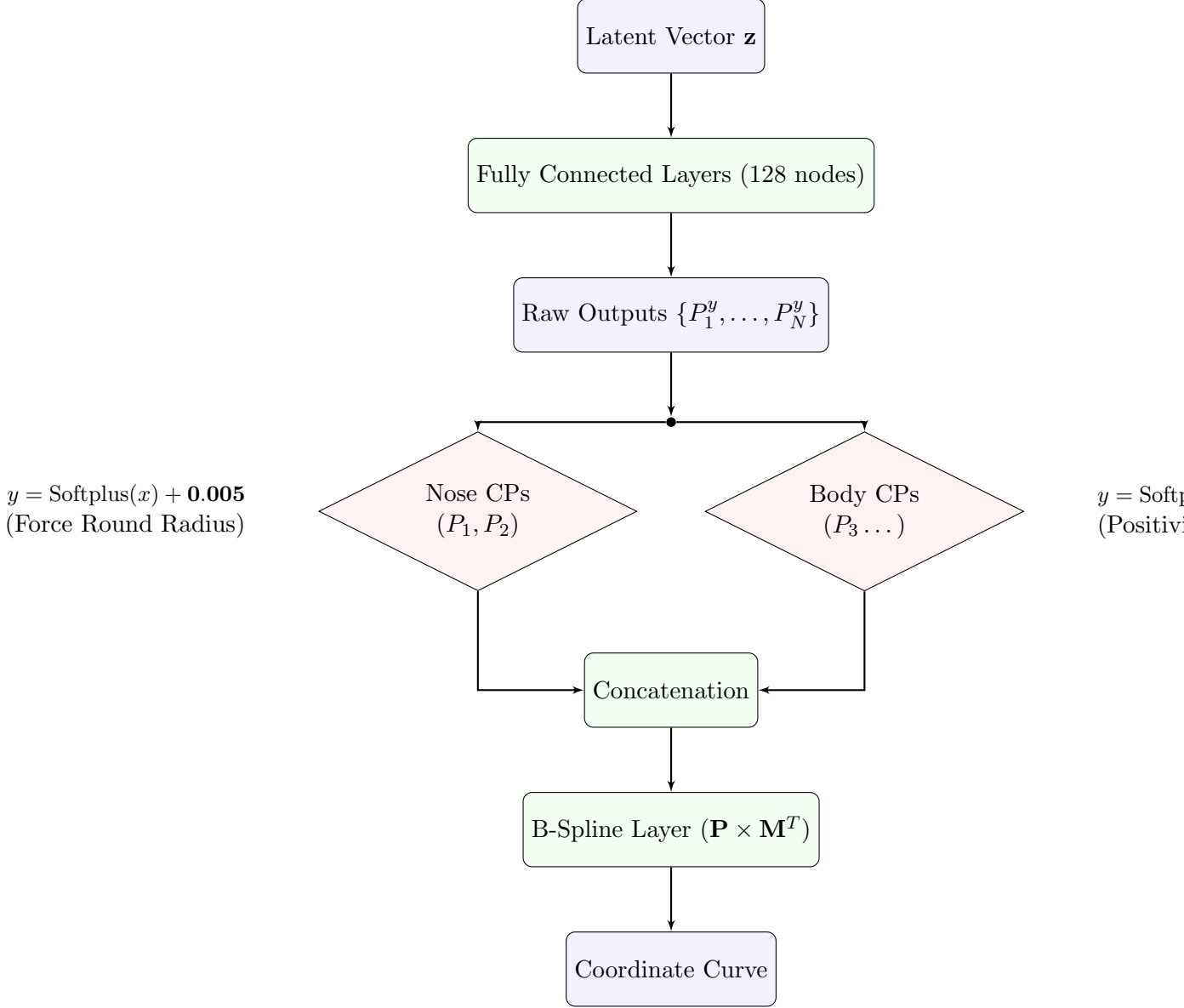


Figure 4: Decoder Scheme with Region-Specific Constraints. A higher bias is applied to the nose control points (P_1, P_2) to prevent the leading edge from collapsing into a sharp wedge.

5 Physics Alignment and Training Strategy

5.1 Embedding Strategy

The latent space $\mathbf{z} \in \mathbb{R}^{12}$ is split to disentangle physical properties from stylistic features.

- **Physical Latents ($\mathbf{z}_{phys} \in \mathbb{R}^3$ per branch):** These dimensions are explicitly trained to track specific aerodynamic features (Max Thickness, Max Camber, LE Radius, TE Direction).
- **Free Latents ($\mathbf{z}_{free} \in \mathbb{R}^3$ per branch):** These dimensions are governed only by the reconstruction and KL terms, allowing them to capture residual geometric variations not described by standard parameters.

We employ a split latent space $\mathbf{z} \in \mathbb{R}^{12}$, divided into **Physical** ($\mathbf{z}_{phys}$) and **Free** ($\mathbf{z}_{free}$) variables.

5.1.1 Physics-Informed Prior (Encoder Side)

Instead of a standard Gaussian prior $\mathcal{N}(0, 1)$, the physical dimensions are trained against a dynamic prior centered on the ground truth labels:

$$\mathcal{L}_{KL} = D_{KL}(q(\mathbf{z}|\mathbf{x}) \parallel \mathcal{N}(\mathbf{y}_{label}, \sigma_{prior}^2)) \quad (4)$$

This explicitly forces the encoder to map the latent variable z_0 directly to the physical value (e.g., max thickness), while the learned prior variance σ_{prior}^2 allows the model to capture aleatoric uncertainty (handling Out-of-Distribution samples gracefully).

5.1.2 Latent Consistency with Gradient Stop (Decoder Side)

To prevent "Posterior Collapse" (where the decoder ignores the physics latent and outputs an average shape), we measure the physics of the *generated* geometry $\phi(\hat{\mathbf{x}})$ and force it to match the latent mean.

Crucial Implementation Detail: We detach the latent mean from the computation graph for this term:

$$\mathcal{L}_{consist} = \|\text{detach}(\mu_z) - \phi(\text{Decode}(\mathbf{z}))\|^2 \quad (5)$$

This ensures that gradients flow **only** to the Decoder (forcing it to fix the shape) and not to the Encoder (preventing the Encoder from "cheating" by shifting the mean to match a bad reconstruction).

5.2 Loss Function and Training

The training objective minimizes a composite loss function comprising four key terms. To ensure stable convergence, we employ a **Normalized Weighting Strategy** where each term is scaled to the same order of magnitude ($O \approx 10^1$) before dynamic annealing is applied.

$$\mathcal{L} = \lambda_{rec}\mathcal{L}_{rec} + \beta(\lambda_{kl}\mathcal{L}_{kl}) + \alpha(\lambda_{phys}\mathcal{L}_{consist}) + \lambda_{reg}\mathcal{L}_{reg} \quad (6)$$

5.2.1 Split & Rebalanced Reconstruction ($\mathcal{L}_{rec}$)

Standard MSE is dominated by thickness errors ($O(10^{-2})$), causing the network to ignore the much smaller camber errors ($O(10^{-4})$). To fix the "flat camber" issue, we split the reconstruction loss and amplify the camber term:

$$\mathcal{L}_{rec} = \sum w_i(y_{thick} - \hat{y}_{thick})^2 + \mathbf{20.0} \times \sum w_i(y_{camber} - \hat{y}_{camber})^2 \quad (7)$$

where w_i is the Kulfan weighting, amplifying leading edge errors ($x < 0.2$) by a factor of 20.0.

5.2.2 Physics Embedding via KL Divergence ($\mathcal{L}_{kl}$)

This term trains the **Encoder**. For the physical latent variables $\mathbf{z}_{phys}$, we replace the standard Gaussian prior with a physics-informed dynamic prior centered on the ground truth labels $\mathbf{y}$:

$$\mathcal{L}_{kl.phys} = D_{KL}(q(\mathbf{z}|\mathbf{x}) \parallel \mathcal{N}(\mathbf{y}, \sigma_{prior}^2)) \quad (8)$$

This forces the encoder to map latent variables directly to physical quantities (e.g., $z_0 \rightarrow \text{Thickness}_{max}$), while the learnable prior variance σ_{prior}^2 allows the model to handle uncertainty. Free variables $\mathbf{z}_{free}$ are regularized against a standard $\mathcal{N}(0, I)$ prior.

5.2.3 Latent Consistency with Gradient Stop ($\mathcal{L}_{consist}$)

This term trains the **Decoder**. We measure the physical properties of the *generated* geometry $\phi(\hat{\mathbf{x}})$ and force them to match the latent mean μ_z . To prevent the encoder from "cheating" (shifting μ_z to match a bad reconstruction), we detach the target:

$$\mathcal{L}_{consist} = \|\text{detach}(\mu_z) - \phi(\text{Decode}(\mathbf{z}))\|^2 \quad (9)$$

This ensures gradients flow **only** to the decoder, forcing it to generate shapes that respect the latent control variables.

5.2.4 Body-Only Smoothness ($\mathcal{L}_{reg}$)

To allow for the high curvature required at the nose, smoothness regularization (second derivative penalty) is applied **only** to control points $P_3 \dots P_N$, explicitly ignoring the nose region (P_0, P_1, P_2).

5.2.5 Scheduling Strategy

- **Cyclical Annealing (β):** The KL weight β cycles linearly from $0.0 \rightarrow 1.0$ every 20 epochs. This periodic "breathing" prevents posterior collapse.
- **Delayed Physics Ramp (α):** The physics consistency weight α is kept at 0.0 for the first 15 epochs ("Geometry First" phase). Once the geometry is stable, α is ramped to 1.0 to enforce physical alignment.